

ActInf Livestream #021.1 ~ John Boik

Livestream | John Boik | 1:48:50

hello and welcome everyone to actinflab

this is active live stream number

21.1

and we're here with john boyk and

several other colleagues looking forward

to a discussion today

on this science-driven societal

transformation series

and project that we've been exploring

and

uh had several background sessions dot

zero one through dot zero four with john

and in the discussion today and next

week we're gonna be looking to draw out

some threads and include some new

perspectives and hear from the questions

that people may have

and for these

discussions we're going to be using the

same jamboards that we were using for

the discussions with just john and i

and we're gonna jump around the

jamboards we're gonna hear questions

from anyone who wants to ask live or in

uh the youtube chat

and we'll have fun so we'll just start

with introductions and then get a quick

little recap and off we go so i'm daniel

i'm a researcher in california and i'll

pass it first to alex

thanks hi everyone i'm alex

i'm a researcher in systems management

school in moscow russia and i pass it to

stephen

thank you hi i'm stephen i'm based in

toronto i do a lot of work with
participatory theatre and community
development
and i'm researching how
social topographies can be used across
scales
of community psychology and other
development practices
and i'm going to pass it over to lee
thanks stephen

uh i'm lee i'm a researcher based at the
university of york and
i'm studying systems transformation and
my supervisor is johan fazee who i think
you've drawn a little bit john in a
couple of these patients
i was

really really happy to uh find someone
else looking at systems transformation
and um active inference at the same time
but although i think you're a lot
further down the line than i am in
thinking about this so really look
forward to this session
cool well welcome john maybe you could
give an introduction and start things
off

uh sure thanks uh for having me and
thanks everybody for joining us
uh my name is john boyk i'm a courtesy
faculty at oregon state university and
the purpose of these talks
is to discuss the series of papers that
i just recently published in the journal
sustainability
the title is science driven societal
transformation parts one two and three
and if you would go we're going to do a

quick recap of some of the material that we've discussed before we'll jump around a little bit just to catch some main highlights and uh of course feel free to jump in with any questions as we go so if you could turn to uh jamboard one slide one yep you got it okay good so this just is the pictures of the different papers the links are done on the lower left so in case anyone wants to check out the papers please do they're all open source frequently freely available and we're going to focus today we'll jump around a little bit and capture the main main topics but we're going to focus on part three design today all right slide two please and as as we go through this if there's more questions if someone wants more materials all of the links are available all the links to all the papers and some other materials are available at my website for this project principled societies project and i might mention there's also a interactive model there that you can delete a framework which we'll talk about so you can see how the economic simulation

goes uh slide five

or slide four excuse me

yes

so what is the what is the purpose of

this what is the what is the problem

that this this series is trying to solve

um the problem that faces

society or really human civilization is

a host of

very large so far intractable problems

including climate change biodiversity

laws poverty

uh income and wealth inequality you know

you know soil soil depletion groundwater

depletion pollution i mean there's it's

a long list of very serious

problems that we face so serious in fact

that it's not clear at this point

how civilization will fare in the coming

decades even in the coming years i mean

the world is armed right now the world

is armed to its teeth with nuclear

weapons

so uh you know

any moment things could go awry or

or

things could just deteriorate

rapidly as climate change and

biodiversity loss gain speed

so hopefully we will solve our problems

hopefully as societies um hopefully we

will improve quality of life and quality

of the environment

uh but that is the per that is what

we're shooting for in the series is

pathways to do just that

i i would like to point out that this

series is focused on transformation in

the sense of the novo design of
fundamentally new systems societal
systems
which i talk about six i believe
so economic systems legal systems
financial systems
governance systems things like that
so the the the problem is the major uh
constellation of problems that we face
uh globally and the proposed solution
solution is to view society as a
cognitive organism
and just like any organism it has a
intrinsic purpose
of maintaining achieving and maintaining
vitality
so how does it do that how does any
organism survive it learns it develops
it learns it acts
it learns from its actions
and and it adapts to new conditions as
necessary
this series is actually a an r d pro
proposal it's it's aimed at the
uh
the global scientific body and my hope
is to interest the scientific world in
this
in this problem of
how do you transform how how could we
build a better society what if we were
to if we were to create
any conceivable way to organize society
any conceivable
economic systems or governance systems
or
legal systems
educational systems what would be the

best in terms of
quality of life and quality of the
environment and long-term
sustainability i think that's a question
that science is now capable of
addressing
perhaps this is the first time in
history that science had the tools and
the skill and the background and the
computational power to address these
kinds of very very large questions but i
think that we can do it now
the whole r d program is is is is
described as a as a
partnership between the global
scientific community and local
communities so
the idea here is to test and to build
create and test new systems at the local
level in in cooperation with local
volunteer communities
[Music]
um
and i and i talk about implementing
those systems and implementing field
tests via a club model which we'll
discuss a little bit more
uh okay so that's a very quick overview
um maybe on to slide five but first uh
stephen with a question and then anyone
else who's raising their hands sure just
keep checking at the hands yeah go ahead
stephen
thank you yeah just uh one question i i
know she mentioned um
architecture so you've got the idea of
architecture and you've got
um

these different things that we have got
in the
constellation
and i wonder if you could speak a little
bit more to
the idea of architecture and
how that relates to architectures in the
brain or maybe distributed systems of
the brain you know yeah you know
the brain is what i'm thinking of when i
use the work architecture well a little
bit i mean people talk about computing
frameworks and computing architectures
too but i really had in mind the way the
brain
functions
and
maybe i should make that even more broad
because if if we
if
if if climate change and biodiversity
loss and these other major problems that
we face
are symptoms of a cognitive dysfunction
at the societal level i mean that's what
i would call it the societal dysfunction
cognitive dysfunction because we've
known about these problems for decades
you know with climate change we've known
about it for
more than 50 years
and yet we have it we've failed to
you know
adequately address the issues
um so i i would say that all of these
major problems are symptoms of
a cognitive problem
or symptoms of a cognitive dysfunction

and then the question becomes okay well
if we're not cognating correctly
what kinds of where do we look for
better models where do we look for ideas
of how we might cognate better and i
would say that in the world of biology
including the human brain there's just
you know a million

or

a large number of examples that we can
learn from

like how does nature compute how
nature's been at this for a very very
long time

and how does it work in nature
and maybe we can

uh align ourselves with the same program
the same process that's going on all
around us

and if we do then there's i would say
there's the chance of
building systems that are much more
effective

at uh at uh achieving the the uh
intrinsic

um you know our goal which is to achieve
and maintain vitality

uh far into the future yes stephen and
then

and i was just gonna sort of building on
that because with active inference one
thing that sometimes comes up in uh
you know is that rather than the brain
being like a computer per se
that is a distributed and actually now
maybe even not even functionally
specialized in certain regions but you
know it's more of a distributed system

of which you know we're not exactly sure
exactly how all old that works but it
with the idea of it being a dynamical
sort of prediction engine so i'm i'm
wondering how
that could
how much might that inform
where you go with this
yeah totally because uh i i we're using
the word brain but but even when i'm
thinking brain i'm thinking whole body
and bodies connected to each other
other bodies in society that
i in in part one of the series
the title is called world view and i try
to set up a context for all of this and
it's part of that context i i i you know
i point out that everything is
intelligent like every piece of life is
connected to every other piece of life
you know either strongly or weakly and
and the whole whole of life is
processing information i mean that's
essentially what life is doing is
processing information
taking reducing the uncertainty about
the past or about the future from the
past
so yes absolutely um you know we have
the four e's for example
uh and we have uh
uh distributed we have
like
in part three i use the analogy of um
between neurons and and individuals in a
society and so how do neurons associate
with other and disassociate with other
there's a kind of a compliment as to how

people associate and disassociate with
other with others in the in the process
of societal cognition
and so we have
connections

in the brain we have connections to our
bodies connections to our emotions
connections to our feet and fingertips
and connections to everything that is
happening around us in our in our
immediate environments connections to
each other and all of that plays a role
in cognition

okay so i'm not yeah so i'm talking
about

cognition in the large sense not in the
narrow sense and i'm certainly not
talking about a computer model that
tells everybody what to do that's that's
not at all what this is about

thanks sean lee and then anyone else who
raises their hand

um it was it's more of an observation
really um it's it's just when you were
saying that you were drawing on the
on the natural biological systems for
inspiration

kind of occurred to me that then what
you're doing is a kind of a a bio
mimicry

yes

but in

but you know building on stephen's point
it's not in the older sense of you know
the mind is computer but much more in
the kind of in your senses you know
biology at all levels being able to
cognite right

right absolutely

i think i read by but mr fuller he said

something one day that he said

i think i'm a process

not a thing

i'm paraphrasing there but that's right

it's like all of life is a process

an intelligent process

cool and and those and those uh just you

know i'll say it again but those ideas

really are new to science and and what

the

i mean you know the mainstream science

new to mainstream science in just the

last

two decades or something like that i

mean fairly recent concepts of

of uh complex systems that uh you know

cognitive systems

uh the interconnectedness of all things

the intelligence of all things trees

communicating with other trees through

you know through soil fungus and

all of these many of these ideas are

actually quite recent in the science

world

and a funny note on that is that

mainstream science is new in the social

cognition world the professionalization

of science

is a new

topic and a new genre and i really like

some of these points about distributed

systems we can think about distributed

systems within one level of analysis or

one system of interest like within the

ant colony it's a distributed system or

within the brain it's a distributed

system but once you start thinking about brains

john you mentioned the 4e the extended inactive embedded and cultured could be more e adjectives but that brings out the richness and says yes it's distributed system amongst the neurons in the brain

and then the brain is interfacing with the body and the body with its niche and the tools and the social affordances and it's starting to sound like a lot of topics that we've talked about on active live streams

so it's so awesome to to see how we're taking some threads from the past and weaving them together and i hope that we can draw this out how a multi-scale distributed systems perspective on cognition coupled with a social transformation agenda

and preference vectors representing policy can get combined through active inference into something that's applied um right right how to act how to make decisions under uncertainty that's what all this all of this is about

as a as a counterpoint to some of that or is a different view on some of what we just talked about imagine if we can look at how

our society treats individuals i mean we can look at

extreme inequalities of wealth and income

uh

different levels of access to healthcare

you know like there's there's all these
issues social issues that we face right
now
imagine if your brain acted that way
somehow you know with with different
neurons where some neurons were just
simply starved some neurons were
excluded from the computational process
some neurons were given very few
resources just barely enough to survive
and others had so many resources that
they were
they were like wasting their resources
because they had so much i mean imagine
how well the brain or how poorly the
brain might
function if that's the way neurons
interacted if that was the community if
if the community of neurons was anything
like the community of individuals in our
society we wouldn't be able to
make hardly any good decisions we'd be a
mess
and we are a mess it reminds me of
dennett's fame in the brain model
and uh the firing rates of neurons are
not identical some are firing many times
per second others fire
not even in a day perhaps
so there's there's a lot
of distributions in these complex
systems
and how do we think about the whole
distribution as an outcome of a complex
process rather than just trying to
identify some sort of an outlier or some
sort of single phenomena it's it's uh
not the leverage point that system

interventions will be effective at so
stephen
and then any other raised hands
this also ties in i think nicely with
the idea of post-normal science um the
idea that in the post-normal science
world where you in high levels of
uncertainty you have to make decisions
so covid is a
classic case right we didn't they had to
make choices in january february march
april even now and you don't have the
data right you don't know
and in the same way with active
inference the brain and
different parts of the body are making
action policy selections
as is
going to inform
action and that's not quite the normal
realm of where science is you're in this
but this is where we're going to be
going as we hit climate change and all
sorts of other complex questions so
uh
that also is quite an interesting tie-in
because um kristen i know if you know
but he did some modelling on kobit 19 as
well yeah he's got into that yeah john
maybe you could um unpack that in the
context of second order science which
you discussed in the papers
yeah or yeah or maybe i can just pass it
to lee and you can say
two words
yeah
i know i'm happy to um
so uh

the

the concept of second order science

really is is just that the the idea that

the scientists can't be separated from

the systems that they are studying so

really where that's kind of led to in

systems transformation research is is

that the only way that we can really

understand systems is by engaging with

them and and learning how to change them

ourselves so we can't kind of stand back

and just try and focus on epistemic

knowledge alone really and that also

implies this kind of second order

understanding of ourselves and how our

own cognitive systems work as part of

the systems that we are engaging with

and yeah that was a value-driven

value-driven science where the or the

scientist is part of the experiment with

the stakeholders

absolutely

while we have you here that paper by

faisy was was authored by a very large

number of people it seemed

and i can you just say if you know the

story behind that uh

uh well a little bit i think it was

authored before i started uh studying

with him but um there's

a transformation community really i

guess of researchers and i think it

probably came out of one of the

conferences

that's been going on so it's a it's sort

of the product of the of a discussion

between

many researchers which is the way that

they kind of prefer to write really so
again like moving away from this idea of
you know the single genius to let's kind
of gather and figure this stuff out
together

right

i i reference uh

paper uh

several times in my part number two so
uh

readers can see how that fits into the
bigger picture here and it's it's uh
funny that that idea that the product of
research is collaborative it's a call
and a phenotype it's a group phenotype
how well this paper is and how many
people cite it and how much impact it
have it's not going to be reduced back
down to a single person
and so

let's take that and then just like you
brought up lee the implications of of
learn by doing is that we have to infer
by acting basically

so then what does it mean for the
scientist who's all in on that to be
studying

climate change on a coast or the
conservation of a forest or sociology or
anthropology

um

very interesting questions that i hope
um a lot of different listeners can
see resonating with their own situation
so stephen and then any other raised
hands

i think just also building on the idea
of this distributed input is and in many

cases it may end up
a lot of the may end up that a lot of
the work is not in science anymore it's
like science informed development or
science informed or even science
informing a model and the model informs
practices which inform
the way that communities can get
involved and now the real agent is
actually the community and we're serving
them so this interesting
shift between
where science fits in and
um can change you know i think that's
quite interesting as well yeah you know
i could have i could have titled the
entire series science influenced the
transformation because that is the
concept i was going for i called it
science driven but the the the
my
my idea there was was similar
and also the historical perspective
which includes many world knowledge
traditions and hundreds and thousands of
years and multiple continents not just
the story of western academia
understanding that that has not been a
steady state there's no stationarity
with peer review again mainstream
science especially the um second half of
the 1900s and further
that's new so science is in
transformation and has always been in
dialogue
from 10 000 years ago
so now given the agency that we have and
who we are and our affordances our

preferences what's the move so we don't have any sort of a locked in phase to resist against that's not to say there's not inertia but it is to say that there's actually a lot of agility in the system the question is what are the leverage points where we can actually make that impact and how my active inference come into play

um you know maybe since we're on the topic maybe we can just switch to us jump ahead to just a little bit if i can find it

or you find the slide and then lee go for it

go ahead lee just to pick up on your um your point there daniel really i mean um i've kind of been hesitant to say it uh in the actual inference kind of you know discord server in the like but i see such a huge amount of emphasis on active influence as it means to um you know to create ai and to improve ai and i can't help i mean

i really spent thinking about how can this knowledge be used to improve you know our own individual cognition and societal cognition human cognition really well there doesn't seem to be um i mean maybe i'm just missing some of the literature on that so if there is any you know point me towards it but then it really seems to be that you know like the whole big assumption is you know we got to put this stuff in ai because that's where all the progress is going to be

et cetera et cetera and then i kind of

wonder is that really a wise goal

i don't know

i think active inference and several

other paths so not exclusive to active

inference it re-thinks artificial part

of artificial intelligence because it's

part of our extended niche and so yes

there's been a lot of focus on applying

active inference to machine learning

into reinforcement learning those types

of areas

and then what we're

talking towards here

is the integration of natural and

artificial intelligence under a broader

extended cognitive systems scope so

humans in the loop or real-time

computers just like every stoplight or

every bank transaction is going to be

every tweet is going to be

designing those interfaces

so that

within each blanket

there's functional cognition not viewed

only from the outside in but as part of

a broader system and maybe we find out

that we had good enough artificial

intelligence so-called algorithms in

2004

and that we just had to hook them up in

a different way

uh just like connecting different parts

of piping could give you an engine

that's more or less efficient maybe it

wasn't the case that we needed some sort

of scaling on a larger data set but we

needed a new way to combine sub

components which is what biology does as

well

so

[Music]

just a few more thoughts on that um

so

this the science world and and

particularly this this

cognitive cognitive framework or concept

of what is cognition i think can be

useful both at the societal level and at

the individual level i mean in the sense

that

if

science can help direct the attention of

the world to

to how how intelligence happens

all around us in nature all around us

every day

how you know how we evolve to

uh

to

you know

expand on the information process and

the information processing

the capacity for information processing

so

it helps it help in any you know it just

helps an individual to understand

a little better or have a different

perspective on

what am i doing what what is what am i

why am i here what am i doing why

why

am i

what am i

you know like to to a degree that these

this these new developments in cognitive

science can help to shed some light on questions like that and at the same time for a society like what are we doing here together what what is our purpose together right and if you know your purpose it's a lot easier to know where to go next how to how to frame things in a way that is suitable for your continued well-being and it's a it's a bright line in some senses between first and second order science in the first order science literature there might be a lot of sort of engineering type appeals to well we'll make the society more efficient or science and foreign policy will just simply be better and once we take that second order turn we start to think we're going to have increased clarity about our values it's not going to be an easy conversation we might need to invent new ways of having the conversation or formalizing our values but we need some sort of path that's going to bring us closer to clarity on our values rather than obscuring it with technical advances so stephen john go ahead and then stephen i was just going to say obscuring it with technical uh details but also obscuring it with with incompatibilities between what the science world is saying and doing and

what say the governance world is saying
and doing or the economic world is
saying and doing obviously if all of
these systems are part of the cognitive
architecture you want them you want to
design them so that they are
integrated with each other so that the
signals

you know travel from one to the next
just as the signals in your body travel
from one

you know one system to the next system
yep stephen and i think this also ties
into

partly what this lab has been talking
about quite a lot in trying to enable
this possibility of making those
transitions without losing the rigor of
the modeling because the modeling is
seen as almost another way to get to
empirical data it's not just quite the
same as science but the challenge you
have as things unlike traditional
science going towards engineering you do
have this ability to embrace effects and
values but the challenge is how to stop
things getting

prematurely rarefied how it's so easy
things get rarified and suddenly
effectively you're just using another
word for system boundary and you're kind
of really doing the same paradigm work
with some different window dressing
right so it's really that's the
challenge i think to to and i don't
think anyone's quite managed to do that
yet but i think that's definitely
we're on the cusp this is this is

happening you know right now this this
struggle of
of what does it mean and how do we do it
and the premature ratification as you
said stephen
it's basically there's a a perfect
investing strategy you know if you had
the time machine you'd know exactly when
to buy the dip sell high
we don't have that we're moving forward
we don't have access to the future so
there's two kinds of confidence errors
there would be being overly confident so
too early
zoning in on a solution that's the
premature reification
and the other fallacy or the other side
of that perfect strategy
would be
waiting too long
and so somewhere between waiting too
long and acting too fast is
that trade-off space that we have from
so many different perspectives with not
just finding the value of explore versus
exploit that's gonna be best but
rethinking the explore and exploit
outcomes as part of a process that might
be doing something like balancing
epistemic and pragmatic value for
example in the free energy principle
so it's a really
awesome um conversation and john
continues
and altering how you measure fitness
relative to those two
those two
differences

well since we've been talking talking
about science the role of science in
this why don't we jump ahead to slide
two

uh or uh

germ board two slide sixteen if you
would yep

and while you oh okay that was fast and
uh well uh we get that rolling uh just
one more thought on second order versus
first order science

you know fizzy's paper is great really i
think an important one

but

it's nude i mean that's it's new to the
science world most of the science world
is very focused on what we would call
first order science where the scientist
is a part is is a is an observer of a
system not engaged with
transformation of that system or
achieving value driven goals in that
system

and uh first order science is fantastic
you know we learn we have learned a
tremendous amount of knowledge gained a
tremendous amount of knowledge from
first order science so

it is it is fantastic for what it is and
you know has to be continued uh
but in addition to that we can also
start to be thinking about what is
second order science how how can the
science world be engaged in a
value-driven effort towards
some better future

and uh i suspect that there are lots of
scientists who are frustrated

who are frustrated because they don't know how to there's not an avenue for them to participate in in really jumping in to make this world a better place intelligent people look at our world and they see the enormous stress that we're under in this moment and are fearful of the of the even larger stress that's a that is likely about to be flawless so i i think that there might be a tremendous amount of frustration in the science world of how do we how do we use our capacities to actually fundamentally make things better as opposed to just passing off of you know them you know sort of little trivializing it a bit here but just passing off a recommendation to some policy makers who then do whatever they do and you know the scientist is then out of the loop so i i i think that this is the time for second order science to flourish this is the this is the opportunity for the science community to really get engaged in these deep and important problems that civilization faces right now and in the in the in the series i'm i'm trying to propose an r d program that is a path for the science world to get involved in these in these deeper

questions

and have an impact a deeper impact on
civilization

so this uh this slide is a is kind of a
summary of uh from my paper of of why
should science

you know why should science care about
this and why should science get involved
and we can just we can talk about any
number of these but i'll just sort of
jump through a little bit just to
highlight a few ideas

the very first one is just this idea of
societal cognition that's that's
you know that's kind of new that's
that's uh there's there's great
potential for scientific discovery just
on that question of how what does a
society cognate how does it cognate uh
in you know in what ways by what
mechanisms by what features
and what with what characteristics i
mean there's you know just as far as
expanding the level of knowledge this
question is really interesting

um a second one

so

we're talking about designing societal
systems and obviously this is a really
complex

issue these are not this is not a simple
how do you build a better can opener
right this is a this is difficult and
complex issues and the science world has
has skills that can be really useful in
in addressing these questions and
framing the framing the movement towards
how do we how do we start how do we test

what are the how do we measure fitness

how do we

what is the goal what is the purpose

what are we doing you know like the

science world would have a lot to say

about uh framing this entire

question of what is societal

transformation and how would we move how

would we move forward with it

i might add too

to this that

um

as i see it at least one of the benefits

of having the science world involved in

this question of societal transformation

is that the science world can provide

kind of object sort of objective and

consistent

metrics and frameworks to help guide the

process

that is

turned around uh if

if this if this if this process of

societal transformation this program of

societal transformation

was merely

based on

what a group

uh like you know the the

um

[Music]

this group votes for this let's do this

and this other group votes for let's do

this and it was just it was just a

popularity contest of whatever was most

popular in the moment that's where

that's where that's where the

transformation is going i don't think

the science world would have any much
interest in that like
popularity contests or that kind of uh
that kind of effort that where's the sun
you know where's the knowledge and that
where's the science
it would be on the other hand i think it
would be very interesting to the science
world if transformation was based on
some kind of principled approach some
kind of
some kind like we can agree uh to
to an extent at least we can agree what
fitness might mean for society
and some of that would be based on
objective measures some of it might be
based on subjective measures but
intelligent people can get together and
and conceptualize what fitness for a
society or for a societal system might
look like you know these are of course
complicated problems deep problems
but if we could do that as a community
if we could if we could
find some way to assess fitness in a
reasonable reasonable way a reasonably
coherent way a reasonably consistent way
then
then we have a path to to move forward
of you know different different groups
could try different ways to reach a high
fitness and all of those experiments
would be adding to the body of knowledge
of how do we do this so that fitness
improves right
i don't mean to to say that that coming
up with fitness scores and fitness
evaluations of new systems would be easy

but it isn't it is a topic that we can
tackle and it's because
that because of
our capacity to
conceptualize fitness and and
formalize fitness in the sense of a
formal assessment i think that makes the
whole proc i'm hoping that makes the
whole project interesting to the science
world because now we're now this is
science now this now this is expanding
knowledge this is using uh first
principles and you know scientific
principles to evaluate systems and and
it's not a popularity contest it's not
uh you know this is this is adding to
our understanding of of of how the world
works and how
intelligence works and how cognition
works you know and expanding it out to
the societal level
thanks i don't know i mean maybe there's
some thoughts on that alone but
yep stephen and then anyone else who
raises their hands
yeah just sort of a couple of questions
around that the idea of fitness
landscapes and how that might be used is
i sense you're you're sort of proposing
that you could use these kind of fitness
landscape i suppose you could even have
descent on free energy landscapes
to try and help inform
policy selection i.e maybe which
research streams you're going to go down
um because at the moment you know we're
kind of
either tied into

what the market says
or some other methods but this could
because this fitness maybe you could
separate out because i could see a
couple of different
ways you could look at that then is it
like an extra field of practice to
inform choosing the policy of which
research field to go down and which
avenues or is it also within the field
is it with the fields connecting to each
other i just wondered how that might be
you know it's all of those because
because everything that we do ought to
be serving our purpose you know i mean i
don't obviously there's great room for
play and you know and exploration and
all that kind of stuff i don't mean to
to uh to
make humans uh you know
two-dimensional or something
humans are complex organisms they have
complex needs and all of those needs
must be addressed right
but
but the fitness i s the concept of
fitness
i see that
i see that as permeating multiple levels
right because because yes we would want
to choose policies that help us and then
and then what does help mean what does
it mean how does it look if it helps us
what is what is that all about right so
in some sense if we just say i want to
help we're already thinking i want to
improve fitness somehow right
but not just policy

but even structural design of what how
what is a system what what what how do
how does society make decisions how do
we communicate what is our communication
systems how do we get input from
everyone into this process of decision
making you know
like like so across the board both in
the structural design of systems
and institutions
and uh
and
even to some degree um
uh uh
social norms right through education
systems like what kinds of social norms
help us
um and then obviously too than in policy
but then even in our individual lives
like it helps i think it might help
individuals to understand that
that
what we want is uh improved quality of
life lower uncertainty how can i make
decisions in my own life
that bring me the things that i that i
think might actually make me feel better
that actually might serve me
uh on a on a
even a superficial and deep level right
thanks john uh lee and then stephen
yeah uh just some um thoughts on on the
the fitness
metrics really so um and this kind of
goes to the
go to the point about the way that
scientists need to engage with local
communities because there's quite a lot

of work that's already starting in in
that area and a lot of it is um
it's based around this idea of place
so
scale has already been taken into
account and this this whole kind of
sense of the emerging bioregional
movement which just essentially means
you know aligning your political and
governance boundaries with the
boundaries of ecosystems
um
and that then naturally has the effect
of it
it partitions things in a way that's
meaningful to to local people
um and then you can start to ask them
you know like what is it that they value
within their place because they're
already kind of connected to it in a way
um uh and i mean
i suppose one thing for me is this idea
of metrics really versus counter factual
simulation because once you start
putting a metric on something you know
like you have a number and then you know
that's something that people can't
really care about whereas you know like
if they can imagine how their you know
their local river system for example uh
wouldn't be polluted by you know sewage
being released from the sewage works or
you know run off from them you know and
they could see you know that then has
this sort of the effect of um
you know of engaging people i think
more in in the
uh in the actions that we would need to

collaboratively take to make these kind
of things happen sure sure other
different people might react in
different ways to some people might be
more you know

really appreciate the modeling aspect
other people might just appreciate the
boots on the ground

local change aspect but i would say
there's room for all of that and uh the
bio regionalism

concepts are great i've been a fan for a
long time and who knows maybe in the
future we'll move more and more towards
that so that communities are really
localized in their environment and
and protecting their environment and all
that kind of stuff

that's great in in some ways in some
ways science has to catch up with
communities of change who have been
working on this for a long time these
these concepts and in other ways science
is already you know

pulling pulling communities

you know helping to save more there's
more there's more there's more we can do
there's a bigger picture we can serve
here

so it's a it'll be a give and take i
think between communities and

there's one more thing is the
bioregionalism idea is great and a lot
of work also has been done uh in
alternatives to gdp

so you know the metrics that are that
people are talking about about you know
considering education and longevity and

disease and all these other things as
as opposed to just a single gdp measure
of
societal quality
so there's lots of work being done on
multiple fields
and to speak to the metrics from an
active inference perspective
let's just say that we wanted a river
that was
uh unpolluted we wanted no toxins in the
river so our preference in the active
inference model would be that the river
is clean that'd be in our preference
vector now traditional reinforcement
learning would look at the level and it
would reinforce policies that reduce
pollution locally and then it would
forget policies that didn't locally
now the question is let's just say you
start decreasing it and all of a sudden
it starts creeping back up your
reinforcement learner is going to be
quite confused so an active inference
we can say that our metric
is actually our uncertainty around our
generative model of the river's
pollution so then
what actions can we take if our
uncertainty starts to get wider
how can we increase our precision
on our generative model of pollution and
then of course concordantly with our
values that it be unpolluted how can we
act to reduce our uncertainty
and so it's the reinforcement economic
maximization mindset pick the metric i
want everyone to read that's the metric

reading well no
the preference is that everyone can read
but the metric in active inference is
our uncertainty around our observations
and it's an extremely
subtle at first but eventually very
far-reaching way to think about these
systems and what fitness is because a
lot of times fitness metrics people are
going to be thinking right like gdp or
write like something that we want to see
but we can actually now tease apart
our preferences
from our affordances from our
uncertainty in our generative model and
several other features of active
inference scale-free models and that is
going to help us have a
multi-stakeholder conversation in a new
way that's great great great great great
great great thank you um
yeah i just add to that i totally i
totally agree with what
daniel just mentioned there because
it's not almost it's not an active
inference per se on its own but it's the
ability of active inference to be used
in this epistemic foraging way
and i think the point that was made
there about reinforcement learning which
we kind of think oh that's some machine
learning method but actually
reinforcement learning is the dominant
approach to development
it's like it's the way humans have
structured in a way their own thinking
it's not exactly reinforcement learning
such a bit but it's it's so constrained

and it's so goal driven um and the only integrating way to overcome it and why thatcher then started saying there is no alternative is the market saying it can't deal with the complexity beyond a certain point so you have to offload it to this market which again you get into that kind of just becomes whatever's the favorite of the day and this gives us a way to get into this nuanced complexity it gives us at least the chance to go somewhere other routes and plausibly deliver on that promise you know i think that's really cool it's right absolutely to give like a spatial metaphor on reinforcement learning versus active inference it's like if there's a mountain peak that we know that we want to get to the reinforcement learner would say okay the goal and the reward is the elevation and so it can get trapped on a local elevation peak but then it might not take a step backwards or down a few to get to the highest peak but again if we do a trajectory of policy inference on getting to the top of a mountain again of course as a preference that we have for ourselves in the future we can ask am i reducing my uncertainty about how i'm going to get there when i'm going to get there and the way that we're going to do it together with potentially a group of hikers who have different capacities

instead of how are we going to get to
the top of a little null and then just
stake out so it's a different
conversation style steven yeah than
anyone else yeah and so building on that
is when you have that come into play
is the ability for the active inference
approaches like you're talking about
is
magnified greatly when you don't just
focus on the one person or the two
people who are like the agents of change
but you say well what about the local
people who have no idea about the goal
they have no ability to elevate they
have no ability to contribute to the
reinforcement learning but they know the
slopes they can tell me the history they
can tell me where people fell to their
death in the past or
where there's
you know all these and that then makes
the place made place-based
community-based contributions much more
plausible um as why they they enable the
scientists and other people to
do exactly what you're talking about
yeah
excellent uh just uh just to point out
um
so we're talking about reducing expected
uncertainty about the future right
think of all the ways that that
think of all the implications of that in
say
you know
design of systems or design of
institutions or actions of humans you

know facing some problem
right so we so one question that
immediately arises is do we have enough
information are we collecting the right
information at the right frequency to
even understand what's going on or to
reduce our uncertainty second is how
good are we at anticipating the future
are we do
can can we do that with some accuracy do
we know what's going to happen next we
have any idea what's going to happen
next how good or how good are our are
how good are our worldview concepts that
lead to anticipation of the future and
how good are our
you know computational models about
what's going to happen next
how good is our information information
sharing capacities how are we
communicating with each other are we
communicating the information between
groups and between individuals such that
we can understand what's going to happen
next and understand what our uncertainty
levels are
uh
you know like i can go on but there's
like this it seems like a simple
proposition at first and then you
realize that proposition has enormous
depth
it implies all kinds of things about how
a society might function and organize
itself
one one thought on the wisdom of crowds
and distributed computation there's a
famous example from early in

quantitative genetics where it was at a world's fair and a crowd was asked you know you put in a raffle ticket you can predict how much some cow uh weighs and then the average of the estimates was extremely accurate although many individuals were of course distantly off the mark and a lot of people take that just um straightforwardly as existence of the wisdom of crowds but there's a few pieces of that scenario that really matter the first is that the people attending may have had experience they may have been working in agricultural settings or have experience with getting feedback on their predictions and secondly the estimate was a bit bounded it wasn't that the cow was one pound or ten thousand pounds there was a an estimator and then it was still a cognitive diversity that contributed to that estimate so you couldn't just look at somebody who overestimates that well get rid of that guy he's wrong he's like that's counterbalanced by someone on the other side of that curve that's how we're getting an estimate so having a distributional and a community perspective on decision making helps us recognize and encourage and foster diversity of opinion and also structure a resilient community so it's something that will be really interesting to see how and where these kinds of memes and ideas pop up

yeah absolutely um
valuing differences of opinion valuing
differences of perspective is one one
thing that you had said there i just
added just add something to that is
uh in our polarized society today we we
you know there's like liberals and
there's conservatives at least in the
u.s
and uh
they seem to be in two different
universes almost right but
liberal and conservative
at their best
is a little bit like
exploration versus exploitation you know
the the necessary balance for any
organism or for any decision-making
process to balance exploration of the
unknown and exploitation of what is
already known what is traditional what
what is our history what what do we know
works
uh you know like at their best liberal
and conservatives are both important
just as exploration and exploitation are
both important and problem solving
and they're relative just like left and
right
when you're spinning around and things
are changing
it's hard to say and that's why we can
pull back
to some real pillars
of biological systems instead of trying
to scaffold further and further into the
political or logistical ether
we can actually

return to our grounding as a biological society so lee i saw your hand raised or maybe i didn't or stephen oh yeah stephen and then anyone else who raised their hands that was quite interesting we made that point though about you know we'd look to explore the unknown and i think that in addition to that um because that's from the perspective of say the scientist or the person in the kind of the knowledge um generating game so to speak that's their profession or whatever but there's also that exploring the known but not knowing that it's known the tacit the so this is where communities can come in is like there's there's like they i realizing that some of that unknown is participants in those environments no it just needs to be surfaced this is where some of the work with the environment side of the active inference the niche and these topographies that i'm quite interested in my side can come in because now they're exploring and revealing stuff and in a way which is networked and sort of not just isolated as single stories or experiences and fragmented so i i think that connects with what you're saying can you give one example of what you're thinking so one example we did a project around

water issues in rural
areas of northern maputo land in in
south africa and
so they had different areas of water
catchment so some people were getting
water from a lake but the lake had
hippos and you know crocodiles so it was
a bit challenging for them in different
ways and um so you've got these
different water catchment areas and it's
like
they had the lived experience of having
to go there having to get the water
having to boil it because if the water's
contaminated you've got to boil it and
there's other um what it means then if
you're then doing that if you've got
someone at home who's got hiv
on top of that so then you know so they
can't walk or they've got to have so all
these questions so what they would do is
they would create a fabric map the size
of the room about all their different
routes to water and then share the
stories from eight different or whatever
numbers of hot spots and then that
became the sort of landscape so in doing
so that that wasn't given the answers
but it was informing another process
maybe a week later which was looking for
innovations looking for options for new
approaches so that's one type of
way that you could like reveal something
from the local landscape which would
never have been available to an engineer
from their own knowledge yeah great
stuff great stuff
um so some of what you're talking about

is what is our narrative what is our
individual narrative and what is our
shared narrative

right like what what is our life what is
our experience of life and what does
what do i as an individual have what
what

knowledge and lived experience can i
offer to this group that would help the
group to eventually function better or
make better decisions right

so

that's a really interesting question to
me

for many reasons but one reason is
how

how

suppose we were to move forward with
this this whole project was moving
forward

how do we communicate with each other
richly how what the i would i would
argue that there is no system no
functional system right now

that a million people or 10 000 people
can richly communicate their lived
experience their their understanding of
life their understanding of how what
might happen if a b and c happen
i don't think we have

we're in need of tools to richly
communicate uh uh our you know
and

i mean we're needed tools yeah

so that's a immediate opportunity for
groups to get involved with is just at
just that little question of how would
we communicate and how would we how

would we

how would we make decisions together in

some kind of collaborative fashion

probably how do 10 000 people do that

once or a million people when you can't

all fit in a room

you know that's the question when you

can't all fit in a room to really

thoroughly discuss these issues and

listen to each other how do we do that

at scale

right so that is

and

i think there could be answers to that

that's not an impossible task that's a

really uh interesting and

doable task if we just put our minds to

it

it's but okay let's relate to regimes of

attention and active inference

perspectives on narratives which were

even our first active livestream number

one was on narrative active inference

because we think it's

crucial and if there's going to be a

social narrative like you had brought up

well that's something that is not just

sharded into a person's brain it's not a

holograph where it's the same in the

person's brain as it is at the social

level things that brains know that

neurons don't it's not like each neuron

is holding a little mini brain

it's actually that there is differences

amongst neurons or if you think about

like star wars or movie it's like each

character

doesn't know the whole omniscient scope

and so we're part of a bigger story
and so it's going to be
interesting to explore what will it feel
like as a person and what will it be
like experientially
with a personal narrative to be part of
a broader societal narrative but those
will not be the same thing
and the desire for those to be the same
thing will lead to system fragility in
many cases
so stephen and then any other comments
yeah i think that that that ability to
know certain things is is a very
important point and i think in active
inference something that i've
increasingly been looking at
is
we've got external states and we've got
hidden states okay in different ways so
there's
from the perspective of the generative
model of a say a water engineer there
will be more like there's hidden states
from everything right from everyone to
some extent okay as well as states that
we can engage in
um and
so there will be this question about how
do you engage the hidden states
in a landscape as well as the
external states of the regime of
attention and where can someone else
who's got a different lived experience
contribute to
opening up and maybe telling us where
they think the meaningful
aspects of their landscape are so that i

think that as well as modeling happening
from

our own computational models and being
able to maybe use some tools that we can
model through

participation in placements place
basically

absolutely absolutely so so this tool i
have in mind

it's uh it's

i want to have my voice heard i i have a
story that is that could be important to
others that could be important to any
particular situation that

or topic that society is trying to
decide upon and i would very much like
my voice to be heard and not just a yes
or no or not just to choose a b or c
way i would i have a rich experience
that i want to communicate

now

you know i another person sitting next
to me might have a model of water
quality or whatever the issue is and
they might have something to say about
what their model

implies

but all of those voices right should be
heard all of those voices should be part
of the cognitive process of a society so
so certainly data driven in you know
model formal models of this and that
sure that's great that's fantastic it's
an extension of our cognitive capacity
and at the same time

i'm

intelligent

and i'm you know i can do things a

computer can't

at least today you know i can do many
things a computer can't

so whatever process whatever
communication system and process that
develops out of this ought to be one
where every voice can be heard can be
registered can be incorporated into
this larger

really cognitive process you're talking
about you know it's like sharing and
we're discussing but that's a cognitive
process right for a group because it's
leading towards some better
understanding of the world
and you know improved decision making
about the world

john if i could uh you mentioned data
driven

so to again sort of distinguish and
contrast with how you're potentially
using data driven versus how it might be
uh implicitly used in other settings
data driven who's being driven us the
data are driving us and our policy oh
the policy is going to be data driven
okay well how about it's human driven
how about the policy and our affordances
and the data are like a car and we're
driving the car do we need a better car
does it have too many wheels where are
we driving who whose turn is it to drive
how are we choosing which music is being
played in our human driven car

not

how are we going to fit into a
data-driven human

mode and it's just interesting that

sometimes what is left unsaid
or unstated is where a lot of those
values get swept under the rug it's data
driven what else do you want to talk
about oh values okay well there's time
for values it's like there has to be
time for values but the data-driven
conversation makes it sound like it's
the beginning and the end
right right today yes yes and i think
that can be expanded greatly
um

right so that's that's when i use the
word richness that's a little bit of
what i'm trying to imply is the richness
of experience the richness of
perspectives the richness of of past
information that cannot that i as an
individual can bring to bear on this
issue whatever the issue is and then on
top of that what what does the data what
data do we have how frequent how
frequently do we collect it how what is
its quality how can it contribute to
this discussion

right so very much
this

it's not that the computer has to decide
what to do it's that human beings have
to decide you know what to do how to
given all this information what do we do
how do we frame this how do we frame our
problem how do we are we even looking at
the right problem are we look are we is
our focus too narrow are we you know or
do we need to broaden our focus to
understand better how all of these
different problems

so very much human and the human in the center

computation

and to connect that to first and higher

order science and i'd be curious what

you think the first order project is

like build a bridge the second order

project is like

given what needs to happen is a bridge

the best solution or maybe we can go

about it a different way and the third

order is well what really needs to be

done here what do we value who are we

those types of third-order questions

which potentially even um are on an

escape trajectory from science because

we're not going to be looking to science

in a narrow sense for the answer to that

third order truly human question but we

do need to expand this discussion from

what is the best way to build the bridge

to

how are we going to go about

fulfilling transporting people from side

to side

or making a landmark for our city

there's different reasons why we might

want to build a bridge and people feel

like a cog in the machine

if they're just being instructed at the

first order

like do this right but when we're

including people in higher orders then

we're part of a multi-scale cognitive

system we're not just a quote cog in a

machine so

then anyone else

yeah i think this um i think it's a

really important point and for me it

touches on um

you know metzinger's uh transparency

opacity

kind of uh you know are we

do we understand the the beliefs and the

frames consciously that we're operating

out of so in the in the bridge metaphor

you know you might say

um you know at the third level the third

order

oh it's it's really about how are we

looking how you know what's the ontology

or the set of metaphors that we've even

coming at this system through and what

effect are they having in shaping

uh you know our well a how we perceive

it and be up then our actions towards it

so you know going right back to the

beginning of this discussion it was

quite interesting when stephen asked you

john about

architecture

you know and i'm kind of thinking is you

know is that um you know is that really

is that a metaphor you know the

architecture of the brain is that is

that a way of looking at a brain or is

it something you know this parallels the

discussions about markov blankets etc

etc you know like are these things that

we're imposing on systems or are they

things that are really there

um i think the architecture one is

really interesting because it kind of

comes from this you know whole

constructivist tradition and i mean i

wonder more whether it's not so much

whether it's about the right
you know the most accurate way of seeing
but what particular sets of affordances
does a model or a way of seeing offers
you know in the face of this particular
problem in this particular niche perhaps
so yeah interesting stuff
thank you lee
um john continue or stephen and then
john feel free to continue
yeah just one last piece i think that
this there's
some areas that
i'm trying to apply these isms now and
to sort of put some buckets i think the
social constructivism is is an important
one to think about maybe science in a
way kind of fits in there in a way as a
human pursuit
pragmatism i think is very well aligned
um with
the kind of
approach of active inference and this
approach here if you take pragmatism
from a you know a human not a humanistic
perspective
well from the third one which is an
ecological perspective so
or any ways of knowing as well not even
perspective so you can drop down into
ways of knowing perspectives
you know pragmatism social
constructivism and ecological
approaches and i think
that
nexus is is quite important
as as a sort of
way to hold some of these ideas

this is you know this is a fantastic conversation and uh and i would say that um exactly the kind of conversation that would be needed to to move such a project forward it would be conversations like this you know between scientists within the communities between communities and touching it on the all these uh you know some some superficial and some very deep and subtle issues and questions of how do we what are we thinking how do we do this how do we how do we measure success what do we want what is our purpose i mean you know we're just maybe starting that conversation here amongst us but i would like to offer to the science world that this could be we know this could be a function in the science world of uh you know a pursuit of study a pursuit of of uh attention and uh if it were i think that the you know the i think that's all that's really necessary actually is to put our minds to this and intelligent people uh communicating amongst themselves and engaging the all interested stakeholders could definitely make progress in this in this endeavor so great great to watch

great to be part of
and
yeah john maybe
if you want to go to another slide but
that's leaves the question how do we in
our dot one video it's kind of an
opening the dot one is where we've had
our background with the dot zeros
we've hopefully read the papers and
digested them written comments written
thoughts to ourselves allowed that to
reverberate and then we open the doors
and we come together in a group
discussion for point one and we also say
hey the doors aren't closed more people
can join this conversation
whether it's next week for point two or
on a broader scale so i'm just wondering
whether you answer now or later who can
be included and how can they be included
in participating
right
well
a few words
um everyone is everyone is welcome in
this conversation uh
in in the in the world and communities
in different uh in different
different types of communities the
science community other communities i
mean this is this is a conversation for
all of us to be had
what do we want what is our purpose what
are we doing how can we how can we
improve
uh how can we reduce uncertainty uh you
know about essential variables you know
to put a check more technically

how to do that i don't know i i'm open
to suggestions of how we make the next
step but you know i sort of put out put
out what i could put out in this series
and i'm hoping that these conversations
might
uh lead to
some
actions that uh engage a larger and
larger sphere of of individuals
and organizations so i'm open if if
there's suggestions i'm open
um could you flip to uh let's see that
would be slide two
uh talk to slide eight
yep go for it
so
we've touched on all these ideas and i
thought maybe this might be a a good
moment to to talk about some of the
practical implications
when in our original uh discussions this
slide was coming after the
after paper one which was all about
world view and and
seeing
life as a cognitive process
and
seeing individuals as as overlapping
uh processes of information flow
and understanding that that when we talk
about
a a
community
improving its
its
quality of life and quality and
environment we're talking about an

extended individual an extended
community that really reaches all the
way to the biosphere ultimately right
we're all connected with other
individuals and all communities are
connected with other communities and all
of life is connected we're connected
with our local ecosystem and with this
new ecosystem so

[Music]

when i when i talk about uh vitality
i'm talking about the vitality of the
whole really that's our purpose is of
human we're intelligent we're humans are
very intelligent we can think a thousand
years into the future of what might
happen if we do something today
and humans can use that capacity to
improve and uh reduce the uncertainty
you know of the whole
uh

quality of life can prove the quality of
the environment and biodiversity and all
that kind of stuff so uh with that kind
of theory uh theoretical aspects in the
past what are some of the practical
implications

like how how would we use any of that
information to actually think about
design for societal systems right
so um so this is a list of something
some items that we have talked about in
previous talks and even touched on today
and when we think about what does this
mean practically

i would like to focus that
to a different question really and that
is how do we how does any individual any

organism solve a problem like what is
necessary to solve a problem whether
you're a bacteria or you're a human's
community right
what do you do
and and on the left are i would say some
of the universal aspects of of cognition
uh for example you have to gather data i
mean you're the intelligent
being is processing information and
information means that some kind of data
or
uh
information has to come from the world
and then it can be processed it has to
remember for example
it has to anticipate the future forecast
the future with some accuracy in order
to survive if if i see a you know if a
lie if i see a lion in my path
uh i might want to accurately forecast
that if i don't get out of the way i
might be the lion's dinner right i mean
anticipation is central to cognition
so so we i've only talked about the
first three in the list there but
already let's we could talk about what
does that mean for a community then
so how does a community gather data for
example
and that might be
you know through sensors through
through
um
through
getting the input of others in the
community about some particular issue
um

requires remember like remembering
requires that that information be kept
in some kind of accessible
form some kind of repository of
information

uh

we can think already of ways that
society does the does these now we we
for example the census bureau
collects information uh
you know annually and
every 10 years collects various kinds of
information

there's there's financial
surveys done every few years
uh some of that information is put in
repositories that that at least
researchers can access for different
reasons

but how

you know how does that how would that
differ how do how does what we do today
how might that differ from
if we were really thinking of
communities as cognitive organisms
how might we improve on any of these
processes right

and i don't know if there's thoughts on
that but i can yeah yeah stephen stephen
go ahead yeah i mean one thing i think
that's interesting is this this actually
points as well though to an interesting
dilemma i don't know if it's a dilemma
but certainly a
challenge

in the sense that

um these artifacts and these artifacts
of knowledge that we have

they're kind of
we have the option in a community
setting to place them in our environment
because they aren't they can be
equilibrium
based artifacts so for instance if i
create a book and i put it on a shelf
it could be their hundred years right
and they'll come back and it's still a
book right
so you've got this ability for that kind
of storage now when i remember something
i i'm with active influence i'm
establishing this kind of
priors an update of priors
um this is
generated on these lower level
sensory flux
pieces of um
sensorium which i have at higher levels
potentially
started to
interpret as what you might call a
signal
but it you get into this interesting
question because it's it's like the
error is kind of the signal in active
inference although it's possibly at
higher levels of thinking
it gets a little bit different because
we've been talking about that with some
of the modelling but there's a there's
an interesting
question around as you scale out
where some of this sense
data in active inference is kind of held
in the kind of collective knowing of a
community

which knows how to interact for instance
with the artifacts

yet you've also got aspects a bit like
i've got my clothes and i've got other
things that kind of they part of me are
they not part of me they're part of the
environment you you there's i think
there's something quite interesting and
i i think that's an area i'm really
interested in exploring is
is teasing that out so i wonder what
your thoughts are on that
uh

well part of part of
i think part of what you're
speaking about might be what i've been
calling richness of information right so
so obviously a a census can collect
information once a year about you know
numbers and
education levels and things like that
that's one that's one stream of
information a valid stream of
information and yet another is very
personal it's what it what is your gut
you know what is your gut telling you
about this situation or this
you know this this constellation of
what's happening today what is your gut
feeling
of what's happening right so how does
the how does a community get you know
process uh sense and process
all of that information together if you
have a you know if we're if we're part
of the community and we're trying to
decide on some issue
it would be good if we heard your gut

you what is your gut telling you about
this is this good or bad or should is
this direction good or bad or you know
even if even if there's no
like data data to back it up what is
your sense
based on you know the the the your your
complexity your wholeness your
like we can't even put a word on it what
you what you are or where that
information comes from maybe but what is
this what is the sense that you're
having and i would like to share my
sentence too
and everyone maybe could share their
sense so how do we how do we do that in
how do we do that when there's a hundred
thousand of us how do we do that
well
yeah
uh to give just a few thoughts there's
there's two sides to the conversation
they're sharing communicating and then
there's really engaging and listening
which is part of the idea of attention
topic so 100 000 people speaking over
each other is zero conversation that's
not yeah yeah that's not conversation
right exactly so how do we how do we do
that
and and i don't think that that's not
you know like it's almost surprising in
a way that
that we we have such enormous technical
capacity in today's society to do
unbelievable things
and yet this very basic question of how
do we as a community of say 10 000 or

100 000

how do we richly share information in a
way that all of us feels heard
that that are that we can express our
our

our kind of like hard knowledge and gut
knowledge

how do we participate together in
describing our situation describing our
fears describing our uncertainty
anticipating what is going to happen
next if we take choice a b or c
you know you it's kind of interesting
that that that that

that tool

doesn't exist

in spite of our technological process
and there might be reasons why that i
mean there might be political reasons
and economic reasons why that tool
doesn't exist because if that tool did
exist

people would want to
participate you know like naturally i
want my story heard i want i want my
community to understand what i know
because maybe it's going to help
somebody

and and i want to hear what what
you know in a in a healthy conversation
in a healthy exchange i really want to
know what you're thinking
and i want to i want to
contextualize all this information in
light of

what our purpose is what our goal is
what are we after what do we want
right

yeah

well can i comment on that because i think this is a really important point and and also though as well as see this is what i find i'm i'm recasting speech as an action policy as an extended cognition into a spatial landscape of semantics right so effectively it's actually words could be seen as in a virtual environment that we're projecting out and what that can kind of open up is

that

when

i'm talking to someone and being heard right which basically means being

words are being heard right

so but there's there's certain things that

i can't communicate through words because they're not in that semantic landscape which is the compression of my a useful tool but the thing with tools is

generally speaking it's not very healthy to relate to tools right it's like tools of things

so one non-verbal work can be really powerful we actually had that with our work within the community who use alternative communication you know we've really pushed that so they it can take them three minutes to do three words because if they're using the head switch and they have to scan through each letter and then hit the head switch and

then
so we'll really about how about it not
being about the words and about being
the look of the eye
and that
building up of
non-verbal
knowing between people which at some
point
will probably be well we actually then
articulate that through fabric
landscapes so then they start painting
the room and do world building and then
you start making the words so there's
there's something about um like you said
there may be something that taught well
we need we need the tools but also
sometimes tools just can't go there so
what do we do where we don't use tools
um and then when do we bring it into a
tool-based environment i think this is a
good point yeah yeah yeah
and i think much of human communication
actually happens non-verbally too if we
can be in the same room and i'm watching
your
facial expressions i'm getting a lot
more information about you and your
intent than if we if i was only hearing
your voice if i was only reading your
words so
sure there's there's nonverbal
communication there's artistic
expression there's
there's poetic expression you know using
words in a different way
uh
and

and
obviously we again we as an individual i
want to be heard and if i'm maybe i'm
expressing myself in words maybe i'm
expressing myself in my how i'm raising
my eyebrows or maybe i'm express maybe i
don't even have words to express myself
but i have an image in my head that i
want to communicate somehow somehow we
we have to be able to communicate richly
uh
in
in mass
and how do we do that and part of the
answer to that might be we communicate
in smaller groups
and that information then is is sort of
propagated up the hierarchy or something
maybe i mean you know the the the the
slate is wide open here so this is the
question to the world
how can we communicate richly in mass
such that our purpose might be fulfilled
you know our our purpose of of
maintaining achieving and maintaining
uh vitality into the future
right so that's the big picture and then
how what are the what are the what are
the many tools or
social norms or
educational processes or
you know
any number of things that lead us to
fulfilling our purpose
you know our intrinsic purpose i should
say you know our biological purpose in a
sense
nice john i hope that

people clip that out or hear your
question and really consider it and what
it makes me think about on this slide
and to return it to active inference
again
is really like the rhetoric
of participation because in a
reward maximization culture
things like communicate it's like but
not now we have decisions to make
whereas when we reframe about reducing
uncertainty on a multi-scale so
individual and shared generative models
it's like
have we communicated enough to even know
each other's generative model because
once i know that your generative model
includes this dietary restriction
i don't need 100 text messages a day i
actually can generate recipes that are
going to be compatible with you so it's
not like we just need to have more
communication or a different kind or
these two people need to talk or some
sort of secret trick it's actually that
we can balance a lot of these sometimes
locally contradictory needs
like
um
choose two and you can find a situation
where they align on this list and you
could find a situation or imagine a
situation where they diverge and the
power comes from being able to have that
goal directness and the epistemic gain
and then potentially through peer
mentorship or through extended cognitive
architecture we could be able to mark

you know where is there a yellow flag
where is there a red flag where do we
need to pull back
and share more
um about one person's generative model
or
how are we working as part of an
integrated unit
that's reducing uncertainty given the
preference vector
rather than
well if we each just make as much money
as possible then the group will have as
much money as possible right
so this is a different way that we can
go about having that conversation and as
hopefully we're drawing out
it's something where everyone's
perspective uh isn't just a uh
group to be formed their perspective is
their contribution and their value yep
yeah good good
we can also look to biology too how does
how does uh the cells of a body
make decisions you know like how does a
liver know what to do and when when to
act
how does the heartbeat know when to
speed up or slow down
there's rich communication that happens
multi-scale
some local some distant through hormones
through messengers there's enormously
rich communication that happens between
tissues and cells in order for all the
pieces to
understand what the you know what what
what is necessary now

right so i mean there's we can we can
this is an open question it's a clean
slate there's a there's a great
potential for the science world and
other communities to move forward of
this question of how do we richly
communicate and make decisions together
how do we do that
and uh and there's lots of biological uh
examples for us to look at to gain
inspiration from
so a question a question from the chat
just before stephen
um from yvonne
uh just about the tools
uh
how many people have used the kinds of
tools that you're suggesting or it
sounded like you suggested that it
didn't exist but what is the preferred
field for a startup from yvonne where
does this tools where is this coming out
of this sort of tool for communication
and
collaboration sure i understand the
question exactly where is it coming from
he asked what is the preferred field to
make a startup in
for this kind of a tool
like would it be
in the field of industrial engineering
or oh
[Music]
so i have
ideas in my head of how we might you
know start to build such tools
um
and

the you know the reason i'm thinking
about it is for this very
this very problem of how to communicate
how do communities communicate how if
this club model were to if we were to
move forward with uh
with studying this club model and
eventually developing a field trial with
a club model how does communication
happen within that club model so that's
that's my interest you know is the
application to this problem of societal
transformation and
and demonstrating new ways of of
cognition at the local level
via clubs
but obviously if you had a tool that
could do this there would be tremendous
opportunities for others to to use the
tool too
this is just one example of
you know thousands you might think of
but
um i used i work in the utility industry
and i've seen how messages get
transferred from you know a caller calls
in and says you know there was a there's
a there's an accident there's a wire
down there's sparks flying there's
children around you know there's some
kind of rich situation happening that
others need to know about
and by the time that message gets to the
person in the field or in the truck uh
you know it might be quite garbled it
might be
it might be not unclear at all what the
problem is or what the dangers are or

what the situation is right so that
would just be one example of how
industry might use a tool for better
communication

uh

and and

you know there would be many many more
examples you could think of so i would
my guess would be that if the science
community moved forward with this
problem of how do you richly communicate
with a large group

that

those tools and the concepts that would
come out of that could be applicable to
a to a wide variety of industries and to
this particular question of how does the
club model work and how do how do
communities cognite

thank you john stephen and then anyone
else with raised hands

yeah thanks i suppose this sort of
reminds me of some of the
principles actually systemic coaching
can be an interesting area like looking
at coaching groups that and the system
it seems to be that

um and i was just wondering if you've
encountered um

the viable systems model stafford beers
viable systems model because i think
that some of the active inference work
can recast that you know because you've
got not just

levels of

regulation on the on the body because
it's based on the i think it's the body
on the nervous system of a frog

um or um

or different nervous systems but
that doesn't necessarily have that
bottom-up teleology that you get from
active inference doesn't it so you have
the kind of the idea that the lower
levels also know something it tends to
be the higher levels of regulating the
lower level so

but uh hey that might be something to
look at as one area that'd be
interesting and

um yeah i just thought i'd mention that
in case you know no no great uh and uh
um

you know those are the kinds of ideas
that would you would want to explore as
you build the kind of tools we've been
discussing and on that on that topic
the goal is not to build one tool that
does all of this right we could have
many tools we could you know different
groups could explore a variety of
approaches and tools and
and uh you know

some depths of simplicity and complexity
and you know there's room for
room for

many discussions on on this topic right
um one other just just to kind of
continue with you the idea of coaching
i've been

i've been around enough folks in my life
to realize that some groups and some
people
are really

skilled at communicating and talking
with others and you know getting

facilitating group conversation and things like that and and many times i've been just so impressed with the ability of certain groups or certain individuals to to kind of facilitate communication and and show by example what healthy communication looks like and uh and and at the same time i watch tv of you know i've seen tv shows of terrible examples of how to communicate uh you know examples of dishonesty examples of manipulation examples of holding back examples of of confusion you know examples of shallow shallow thinking and uh you know social norms part of this whole process would also be to do like the club model it would be to develop social norms that are useful for the community right that are reinforced to healthy healthy behaviors or healthy ways of communicating health healthy skill development that would help the community to better process the information it has right so so that kind of coaching is available too and this is something in the narrative as active inference we talked about when we see representations of communication like they are up weighted or our priors are updated to think that those types of communication are more likely or that they're prevalent or even that they exist at all and then we fulfill that prediction by

reducing our uncertainty about our own
behavior and our own thinking through
other minds of other people's behavior
thinking well what happens when there's
two people and they need to is they want
the cuts in front and then you're not
surprised when they do it you're not
surprised when you do it
but there's also a way to show
representations of that not happening so
that it will not surprise either person
when they both slow down
and that shows how our inter-subjective
level of computation is something that
is influenced by both the bottom up you
know the sensory input and the top down
with our cultural scaffolding
and so
systems in change
need to be able to resonate from the
small to the large
or there's going to be some um
pathologies so stephen and then any
other comments
so this actually could be an interesting
or could inform anyway the types of
questions that could compose to carl
fristen himself you know around some of
these practical implications so we could
have these in mind so i was just
thinking you know do you have sort of
i wonder if you actually have thoughts
about oh
are there holes in this or are there
questions around
where this
whole field of active inference is sort
of going in the sort of global sense and

and whether you've thought oh i wonder

what

what

carl fiston would think about that

anyway it's just a question to put

out

carl if you're listening

feel free to

to join in

no i would love to i would love to you

know we

one of the many ways to start this whole

process off would be to have a

conference where we get together some

folks who have experience in these

fields in a variety of fields who could

offer their thoughts and

cautions and encouragements

whatever they may be well well john um

on june 21st 2021

carl firsten will be joining us for a

little symposium and in another sense we

have a nano symposium

every week but any type of logistics or

organization that are the enabling

architecture is what active lab is there

for

great fantastic

fantastic

um fantastic you know i maybe should say

too we've been talking about active

inference and obviously all of us are

interested in that topic but but you

know it could be that the active

inference 2.0 is called something else

it goes you know like we learn something

in the next decade or whatever and

active inference is no longer

you know like we've moved beyond that
and now we're talking about
some other thing you know what the the
important
aspect of this for us is how does
cognition happen how does
you know how how
biological organisms uh structure
themselves and orient themselves and
organize themselves in order to process
information and you know how does that
work
so whether it's active inference or
whether it's
you know something else
sure everything evolves we learn as we
go so it might be you know next year it
might be a different word
but the concept is how do we
achieve and maintain vitality into the
future you know that's how do we fulfill
our purpose and even today for different
non-english human languages and for
computer languages the word is already
not active inference and so that's why
it's always so rewarding and valuable to
be very clear on what active inference
is which parts are drawing from bayesian
statistics from complex systems from
predictive processing from all of these
other areas instead of thinking that we
need to reinvent the wheel because we
certainly don't we're also just a part
of the cognitive architecture and uh
it'll be a quite interesting ride for
sure so maybe in the last 15 minutes of
this point one john
help us uh via another slide or by

raising another salient question what
can we as listeners and participants
think about in this next week
to um kind of simmer during this week
between our two sessions

okay um

okay um

maybe next week we will talk we'll talk
more about you know we've we've only
covered two slides here we had i think
like 60 slides we could have looked at
we covered two of them remember maybe
three of them

uh but the the uh

the

the goal here was to talk about part
three of the paper which is design
so i think maybe for our listeners maybe
the question of like how would you
actually do this what would
i in parts in part three i

i

spent some time talking about a a
prototype called the what i call the
leader framework local economic direct
democracy association framework and the
emphasis there it is the the concept is
a full integrated systems approach but
the emphasis in the simulation trial
that i published is on economics so just
take that alone what would
what would a fundamentally different
economic system look like how would it
function and how how could it apply to a
club model you know like implemented at
the local level what what what did that
look like

and when i say economic i mean economic

slash financial slash monetary
uh institutional you know like all of
those things wrapped up into the word
economic
what how
how would that look
what are the possibilities
right
we'll go there next maybe great question
one thought is um econ eco
and uh ecosystem and economy it's like
they start with the first three letters
they're so close they're almost on the
right track with an ecological
perspective on economics yet you add one
more letter and it changes from uh
ecology to economics and all of a sudden
it's like never the twain shall meet so
it's an awesome question to be
rethinking how these different
intertwined systems relate just like
intertwined systems in an ecosystem of
like primary producers prey animals
predators renewal etc
right right so so on that you know just
to follow that up just a little more
what is an economic system
you know before we go about designing
new ones what what is the purpose of an
economic system
we ought to ask that question first
before we jump into potential designs
right so what is what what is it
supposed to achieve in the first place
and the i think for some listeners or
for you know the general population at
least if you ask that question the
answer would be like duh an economic

system is about to
you know allocating resources back and
forth you know choosing who gets what
and
and how resources are used
but i would argue that that's not at all
with an economic system you know like
like intrinsically potentially what an
economic system is i would say an
economic system is exactly what we've
been talking about so far today it is it
is a cognitive system it is a it is a
different variety of cognitive system
that allows a community to
cognate in a slightly different way like
a governance system allows its community
to cognate in one way legal system
allows a community to cognate in a
different way and an economic system
allows it to cognate in a different way
so if so if the goal is
functional societal cognition
what does an economic system look like
what one thought on that and using the
bodies intertwined signaling systems
like neurological chemical etc from the
point of view of the ribosome the
protein translating machinery insulin is
just another peptide it's just linking
the amino acids together and it's making
a protein but to the point of view of a
receptor the insulin molecule is in
affordance for binding and then to the
point of view of the downstream
signaling cascades inside the cell we
start to see semantics enter into the
picture and then there's a turnover for
the insulin and there's lactate being

shuttled from the muscles which are
using it and they're they're calling out
for help through these different
mechanisms so it's like
a cargo ship might be carrying
something
and it's putting it in a box and it's
just moving it like an encrypted message
it's just moving the information it
doesn't need the semantics to be able to
relay the information and similarly when
there's these intertwining systems it's
like that one insulin peptide can be
handled by a producer by a transporter
by a
signal detector by a degrading enzyme
and the idea that all of those pieces
are uh
you know you could just change one out
or you could remove one it wouldn't be
the same system as a whole
so stephen and then any other thoughts
i like the hormone um
reference um
as i think that
the the nature of the hormone system
where you you know the hormone gets
released into the blood
let's say the blood and it travels
around the body
it is actually maybe closer to this idea
of a message passing kind of
signal going out there you know
sort of sense in sort of a general way
which is and maybe this is something
that the economic
system does right is
and you can't do that so easily through

necessarily just
dynamical processes with people right
it's like okay you can you can kind of
infer things you can sort of build
things up but okay you set up a pipeline
of value distribution
or at least the information in relation
to value distribution and how that can
be processed and you can release
your hormones which is basically money
or
budgetary
predictions into that
system and it flows right it flows
around it and it gets picked up and it
becomes more like a traditional signal
that can be computed
and but you can't put out the dynamical
system information because it
it's not a thing
it's done it is dynamical so there's
something interesting there i think
around that that that analogy to
that
it's the system it's the system we're
working with to be able to explore how
message passing algorithms
instrumentally used by us as curious and
motivated investigators
allow us to act in the world to reduce
our uncertainty about blood sugar
so following on that the question two
for the public is what is money
you know like in its best sense what is
money
so
we'll try it we'll try to tackle some of
that next week when we meet again yep so

just on a closing
note we heard
two big questions right there at the end
how can we actually do it
and it would be awesome if people could
read especially the third paper in the
series and look at the letter framework
that john has laid
that uh john has proposed and there's a
simulation there and it's a scaffold for
a lot of rich discussion so on the how
can we actually do this we want to hear
many perspectives as well as people's
thoughts on the third section of paper
and then the second question what is
money which is of course a fascinating
question for history anthropology game
theory so many different areas converge
on what is money and it's a system in
flux as we see quite literally day to
day depending on how closely you follow
crypto
right
thank you everybody for joining us has
been fantastic awesome discussion so
everybody who's um
watching live thanks so much for
participating and next week in number
21.2 we'll have a follow-up discussion
with you john so thanks again for your
engagement and for really giving us this
foundation with the multiple precursor
videos because it really helped and i
think this is gonna be a great uh
conversation sequence so
peace out everybody
see you later bye thanks bye thanks
bye-bye

[Music]